Literature Review

Hitting a plateau in the gym, a point where strength stops improving no matter how hard you train is one of the most common frustrations for athletes, and those who enjoy lifting weights. Strength is an essential quality to develop across all athletic disciplines. Suchomel et al. (2016) establishes the importance of muscular strength in determining athletic performance, force generation, and injury resilience, showing the importance of strength in physical activities. In recent years, researchers have used publicly available powerlifting competition data to describe how strength develops across a typical athletic career, to predict an individual lifter's performance, and to identify exactly when in one's athletic career they are most likely to peak. But despite this recent work, one question remains without testing. Once a plateau has begun, what predicts whether a lifter breaks through it or remains stalled? Answering this question could help athletes, or people who simply enjoy lifting weights to know what factors specifically predict whether a lifter breaks through a strength plateau or not.

Hitting strength plateaus is a well documented phenomenon in exercise research. Steele et al. (2022) tracked 14,690 athletes across 6.8 years who engaged in minimal-dose resistance training, and found that strength gains slowed significantly after one to two years, even though total strength continued to climb slowly after that, reaching approximately 50 to 60% above baseline after six years compared to year one. Put simply, the biggest improvements happen early, and progress becomes small and gradual after. But the research notes that it's unclear if this slow in progress can be overcome through different approaches in training. The participants engaged in minimal-dose resistance training, meaning they did the lowest possible amount of exercise to see performance improvements. Perhaps the results would have been different if they increased their volume of exercise. But even so, this literature establishes that plateaus are not

unique to elite athletes, but are a general feature of long-term strength training across all skill levels.

Latella et al. (2024) used the same OpenPowerlifting data I will use to model strength progression by age and weight class. They found a similar pattern to Steele, big gains happen early in year one (7.5-12.5% improvement from baseline) and slow to about 20% increase in strength after a total of 10 years. This confirms the slowdown pattern we see in competitive powerlifting when it comes to gaining strength after long periods of time. However, Latella's study only describes the average curve across the whole population, it does not attempt to predict outcomes for any individual lifter, as Ferrari et al. (2025) does.

Ferrari et al. (2025) built a machine learning model using the same OpenPowerlifting data to predict a lifter's future total (total pounds across three lifts - squat, deadlift, bench) using the athlete's age, bodyweight, initial strength, and time since first competition as inputs. The final dataset included 54,064 observations from 8,907 unique lifters. They found this model to be highly accurate, with an R^2 value of 0.90-0.94. This shows that this powerlifting competition data can be highly useful, and be used to predict things, which will be useful to the predictive model that I build.

Renner et al. (2026) examined IPF-affiliated classic powerlifters with at least 10 completed competitions to find exactly when in their career they peak in overall strength. What they found was that most lifters peak in strength at around age 27, after 4-6 years, near their 10th meet, using age, competition frequency, and weight-class changes as predictors. This study answers when strength plateaus happen, which is different from my question of what happens after a strength plateau.

When it comes to managing a plateau, most coaches and athletes will turn to a practice called “deloading”, where athletes engage in a planned, temporary reduction in training volume, which serves to alleviate fatigue and re-enable progress, typically lasting 5-7 days and occurring every four to six weeks (Bell et al., 2022). By interviewing 18 national and international level powerlifting and bodybuilding coaches, Bell et al. found that volume, not intensity was the primary focus in deloading, with reductions ranging from 25-50% of normal training. Notably, powerlifting coaches made an effort to protect volume on competition-specific lifts (squat, bench, deadlift) while cutting accessory work more aggressively, meaning a deload doesn’t show up as a drop in a lifter’s usual training weight. Most coaches will treat deloading as reactive instead of pre-planned, only doing so if the athlete shows signs of fatigue or stalled progress. Rogerson et al. (2024) also did research into deloading by surveying 246 competitive strength and physique athletes and found nearly identical patterns of deloading. Deloads last an average of 6.4 days, occurring roughly every 5.6 weeks, and were centered on volume reduction instead of weight reduction. Together these studies show that deloading is a widespread, intentional strategy for athletes to overcome strength plateaus, which overlaps with this project's breakthrough question but from a more practical perspective. Neither of these however, capture how this training behavior links to performance outcomes, and whether or not this strategy serves as a reliable predictor as to whether or not an athlete will break out of a plateau. While OpenPowerlifting data does not account for training strategies like this, deloading is still useful to the scope of the research because it can serve as an explanation to gaps in competition.

Two other studies point to reliable predictors for this project’s model. Pearson et al. (2020) examined 1,000 competitive powerlifters over a year and found that competition frequency had a direct, nonlinear effect on strength performance. Male and overall performance

peaked around an athlete's second competition, and female performance peaked around the fifth, and the study suggested an upper limit of approximately four competitions per year after which additional competition does not result in increased strength. This backs up competitive frequency as a reliable predictor which was also used in Renner et al. (2026)'s research. Likewise, Kwan et al. (2022) surveyed 64 elite International Powerlifting Federation (IPF) World Championship competitors and found that 83% of lifters cut weight before competing. This resulted in, on average, a 2.9% reduction in body mass that was consistently associated with negative physiological and psychological effects, including fatigue, anxiety, anger, and a perceived decrease in performance. This suggests that weight fluctuations, another predictor used by Renner et al. (2026) are not a neutral choice when it comes to performance outcomes, and can have a negative effect on whether or not a lifter breaks through a plateau.

Finally, because this project uses logistic regression to model plateau breakthroughs as a binary outcome, it is worth establishing that this method has been used widely when it comes to athlete performance research. Ab Rasid et al. (2024) used logistic regression to classify skateboarders into ability-based tiers using fitness and motor-ability based predictors, correctly classifying the ability of the skateboarders approximately 90% of the time, showing logistic regression can perform well on binary athletic classification problems outside of powerlifting, Xu et al. (2025) found a more direct comparison, testing logistic regression against decision tree and random forest models to predict binary injury risk in professional athletes. Random forest ultimately performed the best, but logistic regression maintained a legitimate baseline throughout. This is an important precedent for this project, as logistic regression's interpretable odds ratios are well suited to a project whose goal is to explain whether or not an athlete breaks through a strength plateau.

All together, these eleven studies show key insights for this area of research. Suchomel highlights the importance of strength across all athletic disciplines, showing how this problem is relevant to sports analytics. Steele establishes that plateaus are normal, and expected in long-term strength training in general. Latella confirms this pattern holds in elite powerlifters and models the general pattern of their strength progression. Ferrari proves that analyzing powerlifting competition data can be useful to predictive modeling, and Renner predicts when a plateau happens, not what happens afterward. Bell, Rogerson, Kwan, and Pearson show strategies that athletes use to try to break through a plateau, even though these are invisible in the data. Ab Rasid and Xu show the usefulness of logistic regression as an appropriate method for modeling binary athletic outcomes. None of these eleven studies test whether a plateau is something a lifter can break through, or what specifically predicts who and who doesn't. This project seeks to fill that gap directly using a logistic regression model built on OpenPowerlifitng data.

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